GPU Saturation Scorer Report

Generated on 2026-09-03 12:21:21

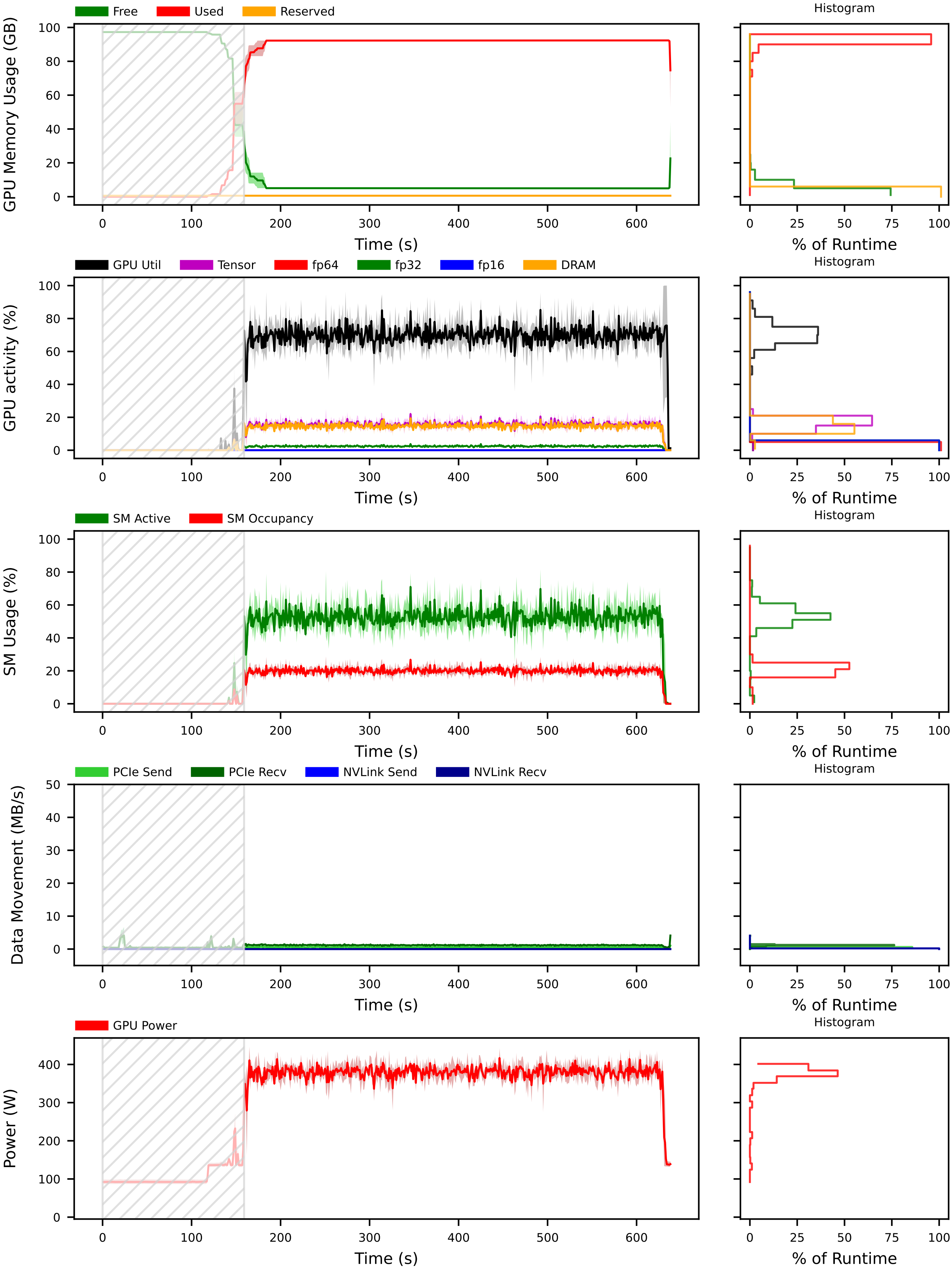
Slurm Job ID	3277344
Job Name	gam-scale
Jobstep	0
Cluster	clariden
Date	2026-09-03T11:52:24+0200
Node Count	1 (of 1)
Task Count	1 (of 1)
GPU Count	4
GPU Energy Use	173.60 Wh
Executable	python
Arguments	'-m' 'torch.distributed.run' '--nproc_per_node=4' '--nnodes=1' '--node_rank=0' '--master_addr=nid007005' '--master_port=36344' 'src/train_robot.py' '--config' 'configs/training/libero_unified/gam/chunk8_150k_2node.yaml' '--results-dir' '/iopsstor/scratch/cscs/veichta/rpr-wp3/runs/ddp_4gpu_3277344/results' '--set' 'training.max_steps=200' '--set' 'training.log_every=10' '--set' 'training.ckpt_every=999999' '--set' 'training.vis_every=999999' '--set' 'training.global_batch_size=48' '--set' 'training.micro_batch_size=3' '--set' 'training.grad_accum_steps=4' '--set' [...]

Evaluation (Experimental)

Criteria	Total Runtime	After GPU Util > 50%
GPU Start ≤ 0 - poor ≤ 25 - improve ≤ 50 - acceptable ≤ 100 - good	75.20 % (good) The GPU usage began very early on during the measurement.	
GPU Util ≤ 25 - poor ≤ 50 - improve ≤ 75 - acceptable ≤ 100 - good	52.58 % (acceptable) The global average GPU utilization could be improved.	56.11 % (acceptable) The global average GPU utilization could be improved.
FP Util ≤ 0.0625 - poor ≤ 0.125 - improve ≤ 0.25 - acceptable ≤ 4 - good	0.13 (acceptable) The global average FP utilization could be improved.	0.14 (acceptable) The global average FP utilization could be improved.
Power ≤ 100 - poor ≤ 200 - improve ≤ 500 - acceptable ≤ 700 - good	309.55 W (acceptable) The global average power draw is reasonable and indicates good GPU activation.	324.17 W (acceptable) The global average power draw is reasonable and indicates good GPU activation.

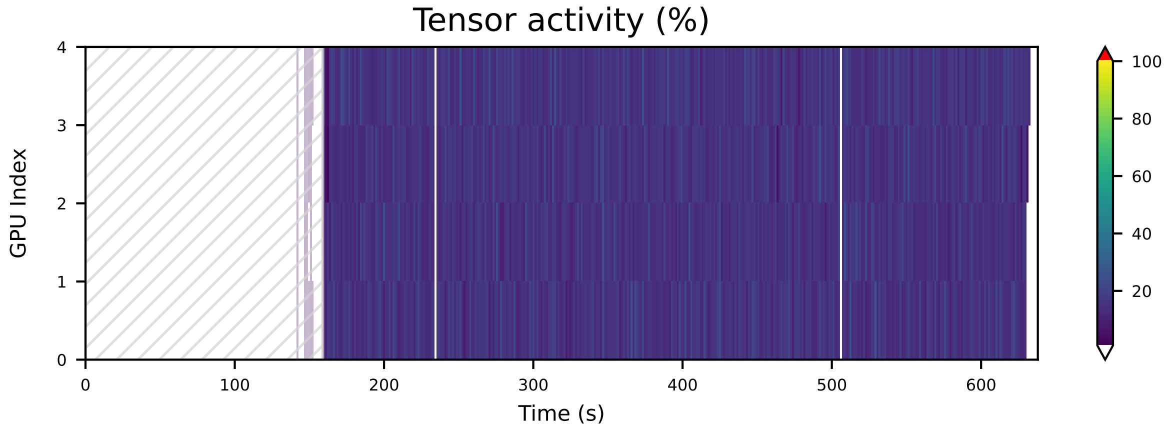
Aggregate GPU Performance Metrics

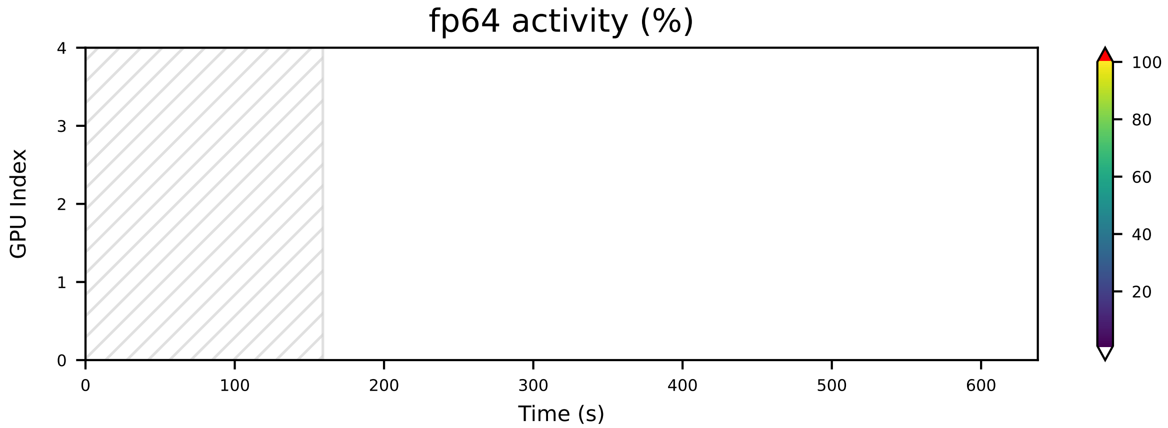
Data summarized over all GPUs.

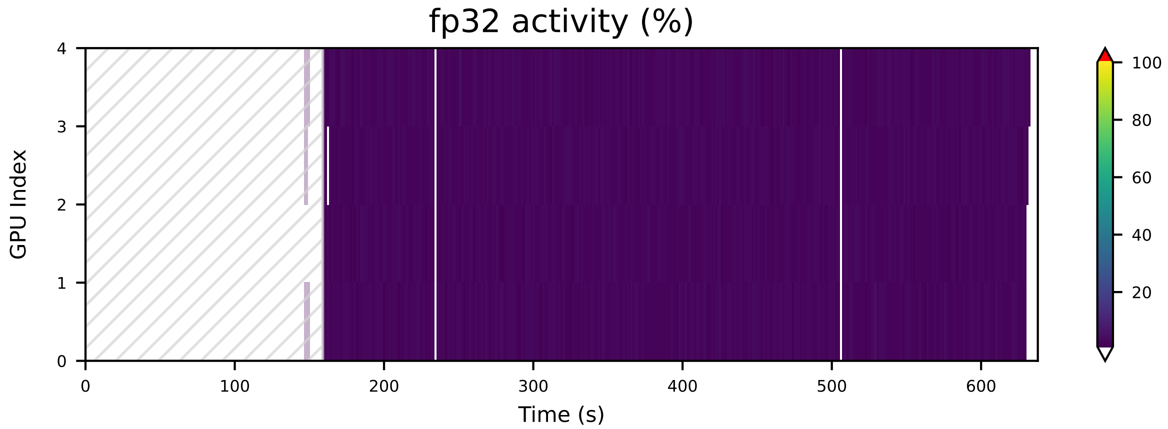


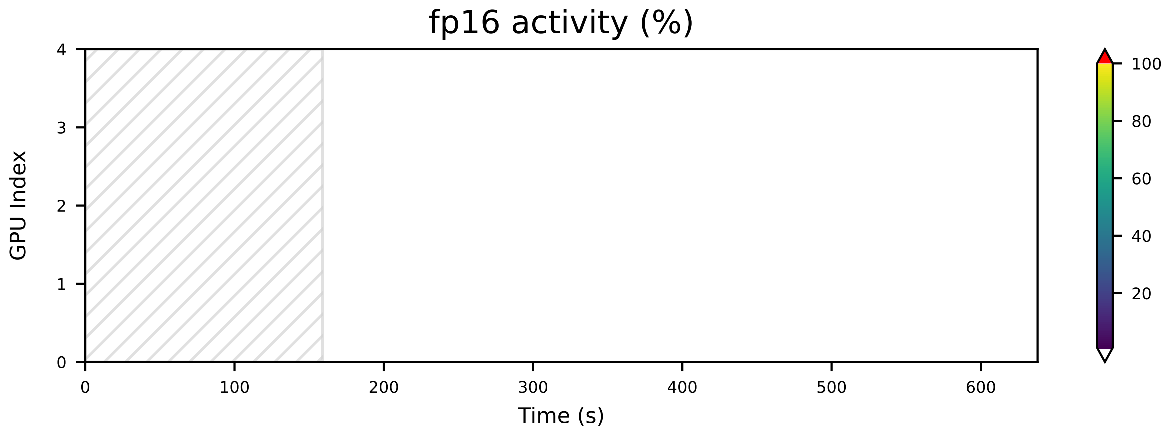
Individual GPU Performance Metrics

Data from all GPUs.









Definitions

From Nvidia Documentation

GPU Memory (MB) Used/Free/Reserved	Used/Free/Reserved Frame Buffer of the GPU in MB
GPU Utilization (%)	The fraction of time any portion of the graphics or compute engines were active. The graphics engine is active if a graphics/compute context is bound and the graphics/compute pipe is busy. The value represents an average over a time interval and is not an instantaneous value.
SM Active (%)	The fraction of time at least one warp was active on a multiprocessor, averaged over all multiprocessors. Note that "active" does not necessarily mean a warp is actively computing. Warps waiting on memory requests are considered active. The value represents an average over a time interval and is not an instantaneous value.
SM Occupancy (%)	The fraction of resident warps on a multiprocessor, relative to themaximum number of concurrent warps supported on a multiprocessor. Thevalue represents an average over a time interval and is not aninstantaneous value. Higher occupancy does not necessarily indicatebetter GPU usage. For GPU memory bandwidth limited workloads, higheroccupancy is indicative of more effective GPU usage. However if theworkload is compute limited, then higher occupancy does notnecessarily correlate with more effective GPU usage.
Tensor Active (%)	The fraction of cycles the tensor (HMMA / IMMA) pipe was active. Thevalue represents an average over a time interval and is not aninstantaneous value. Higher values indicate higher utilization of theTensor Cores. An activity of 100% is equivalent to issuing atensor instruction every other cycle for the entire time interval.
FP Active (%) for FP64, FP32, FP16	The fraction of cycles the respective float-point pipe wasactive. The value represents an average over a time interval and isnot an instantaneous value. Higher values indicate higher utilizationof the floating-point cores.
DRAM Active (%)	The fraction of cycles where data was sent to or received fromdevice memory. The value represents an average over a timeinterval and is not an instantaneous value. Higher values indicatehigher utilization of device memory. An activity of 100% isequivalent to a DRAM instruction every cycle over the entire timeinterval (in practice a peak of ~0.8 (80%) is the maximumachievable).
NVLink Send/Recv Bandwidth (byte/s)	The rate of data transmitted / received over NVLink, notincluding protocol headers. The theoretical maximum NVLink Gen2bandwidth is 25 GB/s per link per direction.
PCIe Send/Recv Bandwidth (byte/s)	The rate of data transmitted / received over the PCIe bus, includingboth protocol headers and data payloads. The theoretical maximum PCIeGen3 bandwidth is 985 MB/s per lane.